

ACTIVITY

IPT 101



INSTRUCTIONS

- You will be creating a simple console-based system for a client with several office locations (Philippines, United States – Eastern Time, and India).
 - Important: Apply the concept of Object-Oriented Programming (OOP) in your code. Use encapsulation, inheritance, and abstract classes for classes that handle mathematical computations.
 - You may search for and use libraries or packages to handle time zone conversion.
 - Programming Language: C#
 - IDE: Visual Studio Community or Visual Studio Code
- The user of the application should be able to:
 - Enter their Employee Number.
 - Enter their Office Location (should only accept the following: Philippines, United States, and India).
- Upon providing the Employee Number and Office Location, the system should record the current date and time, based on the location's time zone.
- (Example: If the employee is based in the Philippines, the recorded date and time should follow the Philippine time zone. If the employee is in India, the recorded date and time should follow India's time zone.)
- All logs should be saved in a dictionary named "Employee Timein-TimeOut Record".



INSTRUCTIONS

- The “Employee Timein-TimeOut Record” dictionary should contain the following information:
 - Employee Number
 - Employee Name
 - Office Location
 - Time In
 - Date: should follow the local date format
 - Time: should be in the twelve-hour format
 - Time Out
 - Date: should follow the local date format
 - Time: should be in the twelve-hour format
 - Total Number of Hours (based on the Time In and Time Out records)
 - Note:
 - If the total number of hours is exactly 9 hours, the note should be empty.
 - If the total number of hours is less than 9 hours, it should contain “early out” and the number of remaining hours before reaching 9 hours.
 - Example: Time in is 8:00 AM and expected time out is 5:00 PM. If the actual time out is 3:00 PM, there is an undertime of 2 hours.
 - Note should contain: Early Out. Hours left: 2 hours
 - If the total number of hours is more than 9 hours, it should contain “overtime” and the number of extra hours worked.
 - Example: Time in is 8:00 AM and expected time out is 5:00 PM. If the actual time out is 8:00 PM, there is an overtime of 3 hours.
 - Note should contain: Overtime. Hours extended: 3 hours



SAMPLE OUTPUT FOR TIME-IN

```
Employee Number:  
1010101  
Which office are you located?  
India  
You clocked in at:  
Date:11/25/25  
Time: 08:56:15 PM  
  
Employee Log:  
Name:John Doe  
Location:India  
Time-In:11/25/25 08:56:15 PM
```

